INCH POUND
MIL-PRF-1/311K
10 July 1998
SUPERSEDING
MIL-E-1/311J
20 July 1976

PERFORMANCE SPECIFICATION SHEET

# ELECTRON TUBE, CATHODE RAY
# TYPES 3ACP1A, 3ACP2A, 3ACP7A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1

DESCRIPTION: Electrostatic deflection and focus.

DIMENSIONS AND PIN CONNECTIONS: See figure 1.

ABSOLUTE RATINGS:

| Parameter: | Ef | Ec1 | Eb1 | Eb2 | Eb3 | ed | Rg | Zd | Ehk | Eb3/Eb2 | Barometric pressure, reduced |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Unit: | V | V dc | V dc | V dc note 1 | V dc | v | Meg | Meg note 2 | V dc | Ratio note3 | mmHG |
| Maximum: | 6.9 | -200 | 1,500 | 2,200 | 6,600 | 550 | 1.5 | 1.0 | ± 180 | 3.0 | 87.0 |
| Minimum: | 5.7 | 0 | - - - | 1,000 | 1,000 | - - - | - - - | - - - | - - - | 1.0 | - - - |
| Test conditions: | 6.3 | Adjust | Focus | 2,000 | 4,000 | - - - | - - - | - - - | - - - | 2.0 | - - - |

GENERAL:

Qualification - Required.


DISTRIBUTION STATEMENT A: Approved for public release, distribution is unlimited.

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TABLE I. Testing and inspection.

| Inspection | Method | Type | Notes | Conditions | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|---|---|
| Qualification inspection | | | | | | | | |
| Pressure (implosion) | 1141 | All | - - - | | - - - | - - - | - - - | - - - |
| Barometric pressure, reduced | 1002 | All | - - - | 87.0 mmHg | - - - | - - - | - - - | - - - |
| Vibration | 5111 | All | - - - | | Width | - - - | 1.0 | mm |
| Direct-interelectrode capacitance | 1331 | All | - - - | k to all | Ck | - - - | 4.2 | pF |
| | | | | g1 to all | Cg1 | - - - | 5.5 | pF |
| | | | | D1 to D2 | C1D2 | - - - | 2.1 | pF |
| | | | | D3 to D4 | C3D4 | - - - | 1.5 | pF |
| | | | | D1 to all | CD1 | - - - | 5.8 | pF |
| | | | | D2 to all | CD2 | - - - | 5.8 | pF |
| | | | | D3 to all | CD3 | - - - | 4.5 | pF |
| | | | | D4 to all | CD4 | - - - | 4.5 | pF |
| Electrode current (anode No. 3) | 5201 | All | - - - | Ec1 = 0 | Ib | 500 | - - - | µA dc |
| Neck and bulb alignment (electrostatic types) | 5101 | All | - - - | | - - - | - - - | 2.25 | inch |
| Cathode illumination | 5216 | All | - - - | | - - - | - - - | - - - | - - - |
| Stray light emission (conventional types) | 5216 | All | - - - | Eb2 = 2,200 V dc; Eb3 = 6,600 V dc | - - - | - - - | - - - | - - - |
| Deflection factor | 5248 | All | - - - | 1D2; Eb3 = Eb2 = 2,000 V dc | DF | 142 | 163 | V dc/in. |
| | | | | 3D4; Eb3 = Eb2 = 2,000 V dc | DF | 110 | 132 | V dc/in. |
| Deflection-factor uniformity | 5248 | All | - - - | | - - - | - - - | 2.0 | % |
| Deflection-factor uniformity | 5248 | All | - - - | Eb3 = Eb2 = 2,000 V dc | - - - | - - - | 3.0 | % |
| Shock | 5115 | All | - - - | | - - - | - - - | - - - | - - - |
| Base material insulating quality | 1216 | All | - - - | | - - - | - - - | - - - | - - - |

See notes at end of table I.



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TABLE I. Testing and inspection - Continued.

| Inspection | Method | Type | Notes | Conditions | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|---|---|
| Conformance inspection, part 1 | | | | | | | | |
| Voltage breakdown | 5201 | All | - - - | | - - - | - - - | - - - | - - - |
| Gas “cross” | 5206 | P1A | 4 | Ib3 = 50 µA dc | - - - | - - - | - - - | - - - |
| | | P2A | 4 | Ib3 = 200 µA dc | - - - | - - - | - - - | - - - |
| | | P7A | 4 | Ib3 = 200 µA dc | - - - | - - - | - - - | - - - |
| Screen and faceplate blemishes | 5106 | All | - - - | | - - - | - - - | - - - | - - - |
| Modulation | 5223 | All | - - - | Ib3 = 25 µA dc | ΔEc1 | - - - | 21 | V dc |
| | | | | Ib3 = 200 µA dc | ΔEc1 | - - - | 45 | V dc |
| Spot position (electrostatic deflection) | 5231 | All | - - - | | - - - | - - - | 10 | mm |
| Spot displacement (leakage) | 5231 | All | - - - | | - - - | - - - | 5 | mm |
| Grid cutoff voltage | 5241 | All | - - - | | Ec1 | -75 | -45 | V dc |
| Pattern distortion | 5103 | All | - - - | | - - - | - - - | 2 | % |
| Grid No. 1 leakage current | 5251 | All | - - - | | - - - | - - - | - - - | - - - |
| Anode No. 1 leakage current | 5251 | All | - - - | | - - - | - - - | 3 | µA dc |
| Anode No. 2 leakage current | 5251 | All | - - - | | - - - | - - - | - - - | - - - |
| Light output | 5251 | P1A | - - - | Ib3 = 100 µA dc | Light | 80 | - - - | FtL |
| Conformance inspection, part 2 | | | | | | | | |
| Heater current | 1301 | All | - | | If | 540 | 660 | mA |
| Electrode current (anode No. 1) | 5201 | All | - | Ib3 = 500 µA dc | Ib1 | -15 | +5 | µA dc |
| Electrode current (cathode) | 5201 | P1A | - | Ib3 = 50 µA dc | Ik | - - - | 125 | µA dc |
| | | P2A | - | Ib3 = 200 µA dc | Ik | - - - | 600 | µA dc |
| | | P7A | - | Ib3 = 200 µA dc | Ik | - - - | 600 | µA dc |
| Base alignment (electrostatic types) | 5101 | All | - | 1D2, pin No. 5 | - - - | - - - | - - - | - - - |

See notes at end of table I.



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TABLE I. Testing and inspection - Continued.

| Inspection | Method | Type | Notes | Conditions | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|---|---|
| Conformance inspection, part 2 -Continued. | | | | | | | | |
| Side terminal alignment (electrostatic types) | 5101 | All | - - - | 1D2 | - - - | - - - | - - - | - - - |
| Side terminal and base alignment | 5101 | All | - - - | Pin No. 5 | - - - | - - - | - - - | - - - |
| Neck and base alignment (electrostatic types) | 5101 | All | - - - | | - - - | - - - | - - - | - - - |
| Base pin solder depth | 1111 | All | - - - | | - - - | - - - | - - - | - - - |
| Screens | 5221 | All | - - - | | - - - | - - - | - - - | - - - |
| Line width “A” (electrostatic deflection) | 5226 | All | - - - | Ib3 = 50 µA dc<br>Ib3 = 200 µA dc | Width<br>Width | - - -<br>- - - | 0.4<br>0.7 | mm<br>mm |
| Angle between traces | 5101 | All | - - - | | - - - | 89 | 91 | Degrees |
| Line width “B” (electrostatic deflection) | 5226 | All | - - - | Ib3 = 50 µA dc<br>Ib3 = 200 µA dc | Width<br>Width | - - -<br>- - - | 0.6<br>0.8 | mm<br>mm |
| Focusing voltage at cutoff | 5246 | All | - - - | | Eb1 | 390 | - - - | V dc |
| Focusing voltage (zero-bias) | 5246 | All | - - - | | Eb1 | - - - | 550 | V dc |
| Deflection factor | 5248 | All | - - - | 1D2 | DF | 175 | 200 | V dc/in. |
| Deflection factor | 5248 | All | - - - | 3D4 | DF | 138 | 158 | V dc/in. |
| Heater-cathode leakage current | 5251 | All | - - - | | Ihk | - - - | 15 | µA dc |
| Secureness of base, cap, or insert | 1101 | All | - - - | | - - - | - - - | - - - | - - - |
| Permanence of marking | 1105 | All | - - - | | - - - | - - - | - - - | - - - |
| Conformance inspection, part 3 | | | | | | | | |
| Life test | - | All | - - - | Group C;<br>Ib3 = 30 µA dc<br>t = 500 hours (min) | - - - | - - - | - - - | - - - |
| Life-test end points | - | | | | | | | |
| Modulation | 5223 | All | - - - | Ib3 = 150 µA dc | ΔEc1 | - - - | 45 | V dc |
| Line width “A” | 5226 | All | - - - | Ib3 = 150 µA dc | Width | - - - | 0.8 | mm |
| Line width “B” | 5226 | All | - - - | Ib3 = 150 µA dc | Width | - - - | 0.9 | mm |

See notes at end of table I.



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NOTES:

1. Accelerator power input (average) should be limited to 6 watts.

2. The deflection electrode circuit resistance should be approximately equal. Higher resistance values up to 5.0 Megohms may be used for low-beam current operation.

3. This tube shall be designed for optimum performance when operating at an Eb3/Eb2 ratio of 2.0. Operation at other ratios of Eb3/Eb2 may result in changes in deflection uniformity and pattern distortion.

4. This test to be performed at the conclusion of the holding period.



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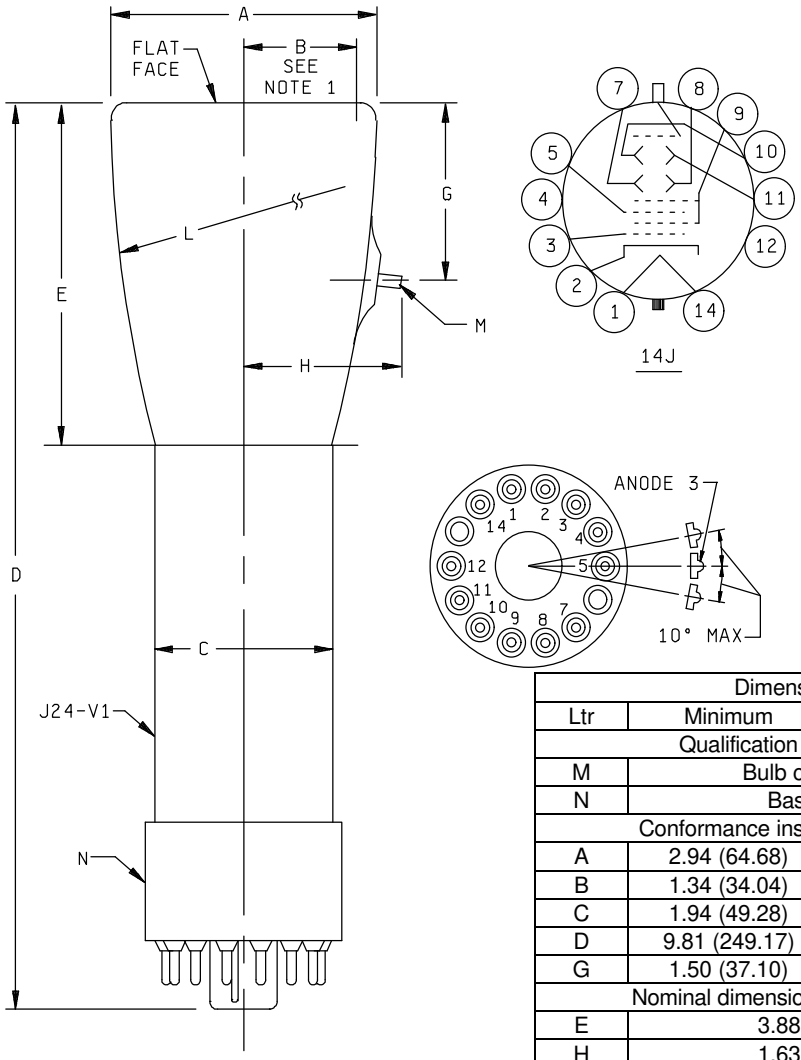


| Dimensions | | |
|---|---|---|
| Ltr | Minimum | Maximum |
| Qualification Inspection | | |
| M | Bulb contact: J1-22 | |
| N | Base: B12-37 | |
| Conformance inspection, part 2 | | |
| A | 2.94 (64.68) | 3.06 (77.72) |
| B | 1.34 (34.04) | |
| C | 1.94 (49.28) | 2.06 (52.32) |
| D | 9.81 (249.17) | 10.19 (258.83) |
| G | 1.50 (37.10) | 2.00 (50.80) |
| Nominal dimensions (see note 2) | | |
| E | 3.88 (98.55) | |
| H | 1.63 (41.40) | |
| L | 12.44 (315.98) R | |

| Pin connections | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Pin No. | 1 | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 11 | 12 | 14 |
| Elements | h | k | g1 | int con | a1 | D3 | D4 | a2 | D2 | D1 | nc | h |

NOTES:

1. Minimum useful screen radius.
2. Nominal dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube types 3ACP1A, 3ACP2A, 3ACP7A.



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Custodians:
Army - CR
Navy - EC
Air Force - 85

Preparing activity:
DLA - CC

(Project 5960-3456)

Review activities:
Army - AV, CR4
Navy - AS, CG, MC, OS
Air Force - 11, 17, 80, 99



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